Total Station

Category: **Measuring & Marking Tools** Type: **Full SOP** Priority: **Medium** 1 labs

Covers these lab items: Total station with prism and prism pole

What it is used for

A total station measures horizontal angles, vertical angles, and distances to determine positions and elevations in surveying. Use the instrument with a correctly positioned prism and prism pole to collect accurate field data.

Main parts

- Total station with telescope and display
- Tripod and mounting screw
- Tribrach and levelling screws
- Optical or laser plummet
- Prism reflector and prism holder
- Prism pole with bubble level
- Battery, charger, and data storage device

Before you start

1. Inspect the total station, tripod, prism, pole, and accessories for damage.
2. Choose firm ground with a clear line of sight to the survey points.
3. Set up the tripod securely and adjust its height for comfortable operation.
4. Mount the total station and centre it over the survey station.
5. Level the instrument using the levelling screws and electronic or circular level.
6. Check battery charge, date, time, units, and required survey settings.
7. Measure or confirm the instrument height and enter the correct station coordinates.
8. Check the prism constant and confirm the prism pole bubble is working.

How to use

1. Switch on the total station and select the appropriate survey job.
2. Enter the station coordinates, instrument height, and required reference information.
3. Sight the known backsight point and set the horizontal orientation as instructed.
4. Place the prism pole vertically over the target point using its bubble level.
5. Aim the telescope at the prism and focus the target clearly.
6. Use the correct prism mode and measure the distance and angles.
7. Record the point number, prism height, coordinates, and relevant field notes.
8. Repeat observations where required and verify results against known control points.
9. Save the survey data and back it up using the approved transfer method.
10. Power off the instrument, remove it carefully, and pack all components in their cases.

⚠ Safety

- Set up the tripod away from traffic, unstable edges, and moving equipment.
- Use high-visibility clothing and warning markers when surveying near roads.
- Never look directly into the sun through the telescope.
- Secure the tripod legs and hold the prism pole firmly in windy conditions.
- Do not leave the instrument unattended in public or exposed locations.
- Protect the instrument from rain, dust, impact, and sudden temperature changes.

Care & storage

- Clean lenses with approved lens tissue and suitable cleaning materials only.
- Keep the instrument and prism dry before storing them in their cases.
- Charge batteries with the recommended charger and avoid deep discharge.
- Check calibration, collimation, and levelling accuracy at recommended intervals.

Common mistakes to avoid

- Setting up the instrument without accurate centring or levelling.
- Entering incorrect station coordinates, instrument height, or prism height.
- Using the wrong prism constant, target mode, or measurement units.
- Holding the prism pole tilted or ignoring its bubble level.
- Saving incorrect point names or failing to back up survey data.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

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